Vienna Behavioral Circle

Automated Corruption Measurement and Electoral Cycles

Evidence from Brazil

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Audits as tool for detecting and preventing corruption

Audit programs reduce corruption and improve electoral accountability

The literature runs on hand-coded measures:

- Resource-intensive covers only **a fraction** of all rounds
- Human judgment under uncertainty noise

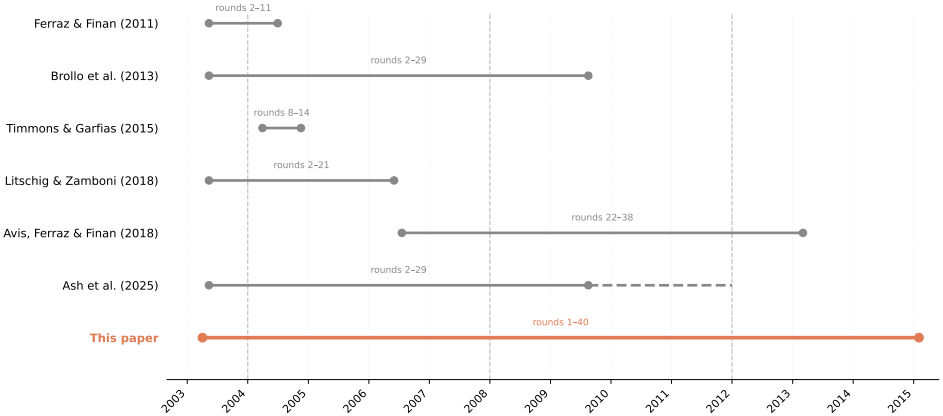
LLMs? Performance degrades on long documents (Liu et al. 2024 "lost in the middle")

Pairwise correlations among hand-coded measures

Measure pair	r
FF(2011) – GT(2015)	0.584
FF(2011) – Brollo(2013)	0.483
GT(2015) – Brollo(2013)	0.320
Brollo – LZ(2018)	0.217

Humans Disagree: Substantial divergence in coding rules and coverage

Audit Related Corruption Measures in Brazil



The Automated Index: Dictionary + PCA

Step 1 Classify each irregularity

Domain-specific dictionary: tokens and n-grams
"fraud", "ghost firm", "lack of proof" vs. *"missing cadastral info", "management report"*

Step 2 Build 5 features per report

- # pages, # lines, # images
- # total irregularities
- # severe irregularities

Step 3 PCA over the 5 features

Why PCA works here

- Pairwise correlations: **0.56 - 0.96**
- PC1 eigenvalue: $\lambda_1 = 3.98$
- PC1 explains **80%** of variance
- All loadings 0.39 - 0.48
  no single variable dominates

Supervised alternatives (NB, LR) correlate at $r > 0.98$ with dictionary-PCA index

Four Validation Tests

1. Human coders

Where human coders agree exactly:
$R^2 \approx 0.70$ (with state FE)

2. GPT-5.3 labels

15,000 irregularities classified
100% agreement with dictionary

LLMs not needed for short text classification

3. Correlates with variables traditionally associated with corruption

Media presence, GDP, Literacy Rate, Urban Population, Distance from the capital

4. Ash, Galletta & Giommoni (2025)

Their XGBoost framework predicts corruption
from budget data. Substitute my index as outcome:

	AGG	**This paper**
AUC (bootstrap)	0.755	**0.861**
AUC (tuned)	0.781	**0.875**
F1 (bootstrap)	0.613	**0.912**

Same features, same code better outcome variable

Full Universe Coverage — What This Enables

Prior measures: snapshots

- FF (2011): rounds 2 - 11
- Brollo (2013): rounds 2 - 29
- Avis et al. (2018): rounds 22 - 38
- AGG (2025): rounds 2 - 29

This paper: rounds 1 - 40, years 2003 - 2015

3 - 5× more municipalities than any hand-coded dataset

Temporal dynamics become estimable
not possible with snapshots

What full coverage gives us

- Track corruption *within* the electoral cycle
- Sufficient power to test heterogeneity (party, term status)
- Reconcile apparently conflicting findings in the literature

Index also used in Muço (2025, JEBO)

The Electoral Cycle in Detected Corruption

Specification:

$$\text{Corruption}_{it} = \sum_{k=1}^{3} \beta_k \cdot \mathbf{1}[k \text{ yrs to election}] + X_{it} + \alpha_s + \gamma_t + \varepsilon_{it}$$

Baseline: audits > 3 years before election

State and year FE; clustered SEs at municipality

Results:

- **1 year out**: -0.257 SD $(p < 0.01)$
- **2 years out**: -0.209 SD $(p < 0.05)$
- 3 years out: indistinguishable from baseline

Interpretation

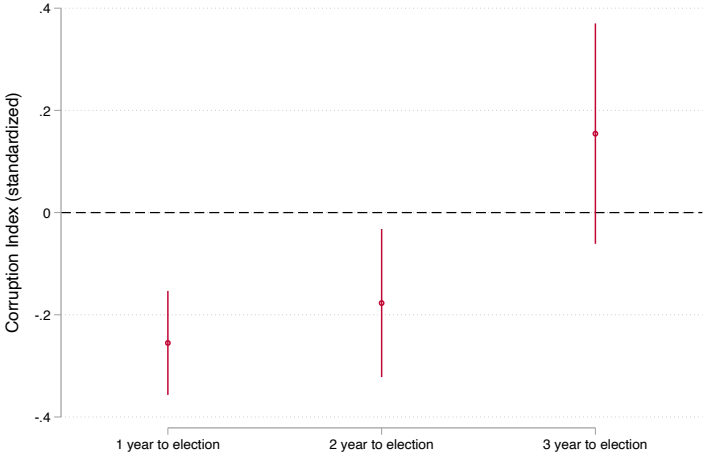
Audits deter, but *temporarily*.

The "shelf-life" of deterrence: $\approx$ **2 years** concentrated pre-audit.

Consistent with Bobonis et al. (2016) on strategic electoral timing.

Extends and qualifies Avis et al. (2018) on persistence.

Visualizing the Electoral Cycle



Who Drives the Cycle? Ruling Out Obvious Mechanisms

Mechanism 1: individual re-election incentives?
First-term (re-electable) vs. lame duck mayors

→ Not driven by personal career concerns

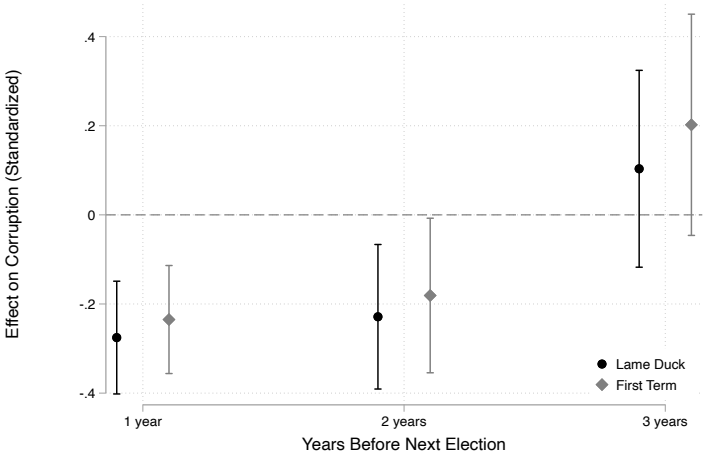
Mechanism 2: partisan enforcement?
PT (presidential party) vs. PSDB (opposition)

→ No evidence of differential enforcement by party

What remains

The cycle is **broad-based**.

Operates through *systemic* election-period pressures party-level, institutional, or broader political channels.

Auditor-driven explanations also difficult to sustain:
if auditors intensified scrutiny pre-election, corruption should *increase*, not decrease

First Term vs Lame Duck